A User-Controlled VR Coping Hub: Three Interaction Cases

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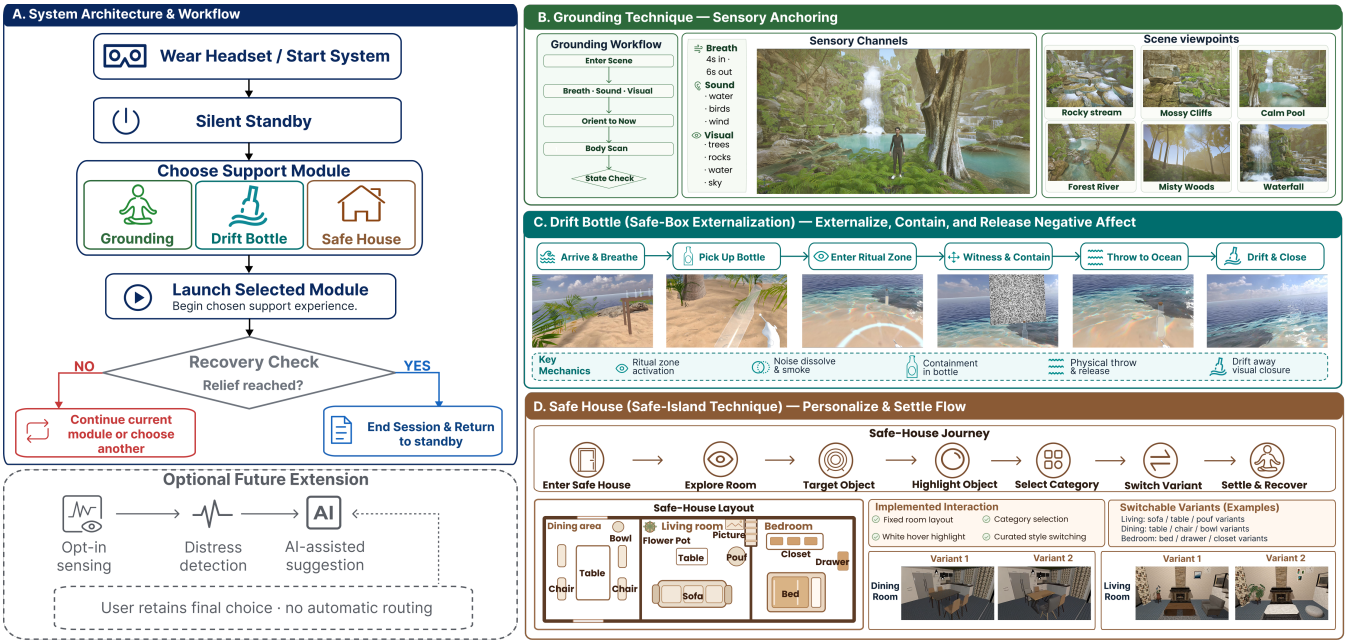


Figure 1: System architecture and workflow. Users select a module, enter it, complete or exit, and return to the shared launcher; each module then follows a distinct interaction grammar. The sensing extension is a future pathway.

Abstract

We present a user-controlled Meta Quest 3 Hub for three coping-oriented interaction cases: Grounding, Drift Bottle, and Safe House. Users manually select, enter, exit, and return from each module through a shared lifecycle. The cases instantiate sensory anchoring, metaphorical externalization, and reversible environmental personalization. A formative Hub evaluation (N=36) and separate Grounding pilot (N=20; 2×2; n=5/cell) provide descriptive evidence. Our contribution is a control architecture; we claim neither clinical efficacy nor automated routing capability.

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VRST '26, Sendai, Japan

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ACM ISBN 979-8-4007-2811-2/2026/11
https://doi.org/10.1145/3822517.3849892

CCS Concepts

• **Human-centered computing** → **Virtual reality**; *Systems and tools for interaction design.*

ACM Reference Format:
Qilei Sun, Ruoxuan Mai, Yeyun Wan, and Binwei Lei. 2026. A User-Controlled VR Coping Hub: Three Interaction Cases. In *32nd ACM Symposium on Virtual Reality Software and Technology (VRST '26), November 16–18, 2026, Sendai, Japan.* ACM, New York, NY, USA, 2 pages. https://doi.org/10.1145/3822517.3849892

1 Introduction

We ask whether guided grounding, metaphorical externalization, and personalized environments can remain jointly available while users control selection, entry, exit, and return. Prior XR coping systems have explored these mechanisms; here, we combine them in a shared user-controlled launcher. Our architectural contribution is a shared launcher for three selectable interaction grammars, without

automatic recommendation, routing, or forced progression. We report descriptive lifecycle evidence from three working prototypes, not clinical efficacy.

2 Related Work

Prior HCI/XR research explores coping-focused XR mechanisms in varied forms: mixed-reality emotion regulation [3], metaphorical distress externalization [2], and restorative safe-place systems [5, 6]. While these works inspire our module designs, we focus on enabling heterogeneous mechanisms under a shared select-enter-complete/exit-return lifecycle. Our contribution lies in this user-controlled orchestration of diverse interaction grammars, not novelty for the individual coping mechanisms themselves.

3 System and Interaction Cases

The Hub's shared-launcher navigation lets users manually select, enter, complete or voluntarily exit one of three modules, then return to the launcher for further selection. There is no automatic recommendation, routing, or forced progression. Under this unified lifecycle, each module implements a distinct interaction grammar. Grounding: *orient-breathe-sense-check* for body awareness, situational orientation, sensory noting, paced breathing and state check. Drift Bottle: *grasp-contain-throw-observe*, collecting and discarding visual distress via a drifting bottle. Safe House: *point-hover-select-switch-confirm* to browse rooms, assess layout variants, and pick reversible environment configurations.

4 Evidence and Evaluation

The Hub formative evaluation tested three modules (N=36); a separate Grounding pilot yielded module-specific exploratory results (N=20). Participants were randomly allocated to six counterbalanced module orders (n=6 per order): G–D–S, G–S–D, D–G–S, D–S–G, S–G–D, S–D–G (G=Grounding, D=Drift Bottle, S=Safe House). Logs tracked entry, completion, error, help, exit, and return. Post-module 1–7 ratings (higher = more) measured clarity, perceived control, and reuse intention; task time was logged in seconds. Figure 2 presents the rating and task-time metrics; return-to-Hub success is reported below. Return-to-Hub success reached

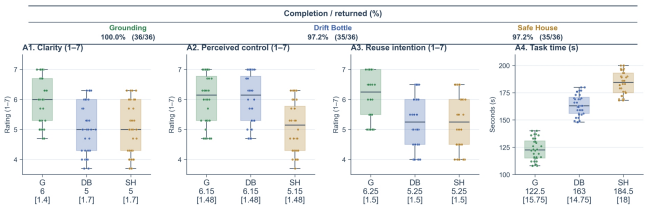


Figure 2: Hub formative evaluation (N=36). Points are participant-level module trials; boxes show IQR and median. Clarity, perceived control and reuse intention use 1–7 ratings; task time is in seconds.

35/36 (Drift Bottle), 35/36 (Safe House), and 36/36 (Grounding). Grounding had the shortest median task duration and the highest median clarity and reuse intention; Grounding and Drift Bottle scored highest for perceived control. Full distributions and summaries appear in Figure 2.

Separate Grounding Pilot. This pilot adopted a randomized 2 × 2 between-subjects design (N=20; n=5/cell), factoring Dialogue Intelligence (LLM-dynamic / scripted guidance) and Visual Embodiment (avatar / voice-only): G1(LLM+avatar), G2(script+avatar), G3(LLM+voice-only), G4(script+voice-only). Stress Recovery Score was post-stress minus post-intervention adapted STAI-S [4]; higher values indicate a larger decrease in this STAI-S score. Perceived restorativeness was measured using a four-item adapted Perceived Restorativeness Scale (PRS) [1]; scores were the means of 7-point items, with higher values indicating greater perceived restorativeness. Group means for Recovery: 22.20, 15.00, 9.60, 15.20; for PRS: 6.00, 5.45, 4.50, 5.45 (see Figure 3). Outcomes are exploratory and module-specific, not Hub-level efficacy evidence. Ethics approval: ER-SRR-157821666202602242140 08; informed consent was obtained from all participants.

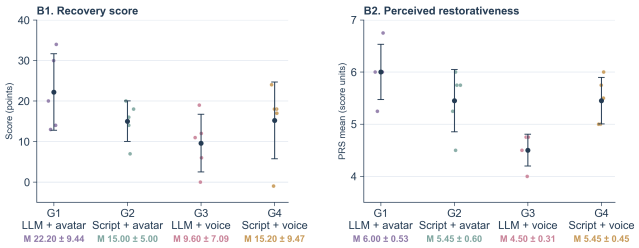


Figure 3: Grounding pilot (N=20; four cells; n=5/cell). Small points show observations; large points and bars show condition means ±1 SD. B1 reports Stress Recovery Score points; B2 reports PRS mean score units.

5 Discussion and Conclusion

The Hub is a user-controlled architecture that keeps three interaction grammars jointly available for manual selection, exit, and return. Our Hub study targets lifecycle usability; the Grounding pilot yields only module-level exploratory evidence, not Hub-wide or clinical efficacy. Automated recommendation and adaptive routing remain future work.

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